O-A-76E
July 3, 1984
SUPERSEDING
O-A-76d
May 9, 1968

FEDERAL SPECIFICATION

ACETIC ACID, GLACIAL, TECHNICAL

This specification is approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers one technical grade of glacial acetic acid.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-186	-	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals
PPP-C-2020	-	Chemicals, Liquid, Dry, and Paste: Packaging Of
PPP-D-729	-	Drums, Shipping and Storage, Steel, 55-Gallon (208 Liters)
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

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(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Washington, DC; Philadelphia, PA; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Code of Federal Regulations (CFR)

49 CFR 171 to 179 - Hazardous Materials Regulations

(The Code of Federal Regulations and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM Standards:

D 1193 - Reagent Water
E 1 - ASTM Thermometers

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Appearance. Glacial acetic acid shall be a clear, colorless liquid at temperatures above its freezing point and shall be free from sediment or suspended matter when tested as specified in 4.2.4.1.

3.2 Dilution. No turbidity shall develop when the glacial acetic acid is tested as specified in 4.2.4.2.

3.3 Assay. Glacial acetic acid shall have a freezing point of no less than 15.7°C when tested as specified in 4.2.4.3. (This is equivalent to no less than 99.5 percent by weight as CH_3COOH.)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the glacial acetic acid produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105, inspection level S-2.

4.2.2.2 For glacial acetic acid test. See 6.5 for sampling and testing precautions. Sampling shall be conducted in accordance with table I. A representative specimen of approximately 250 milliliters (mL) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

4.2.2.3 For container leakage test. Sampling shall be conducted in accordance with MIL-STD-105, inspection level S-2.

TABLE I. Sampling for glacial acetic acid test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample unit, intermediate, and shipping containers shall be examined for the following defects using an AQL of 1.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Cleaning, phosphatizing, and painting not as specified (when required)

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(f) Fiberboard pads missing or not as specified (when required)
(g) Strapping or taping missing or not as specified (see 5.2.2.1)
(h) Marking incorrect, missing or illegible
(i) Unitization not as specified

4.2.3.2 For glacial acetic acid test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.3.3 For container leakage test. See 6.5 for sampling and testing precautions. The sample unit shall be one container. The sample containers selected in 4.2.2.3 shall be tested as specified in 4.2.5 using an AQL of 1.5 percent defective.

4.2.4 Glacial acetic acid tests. See 6.5 for sampling and testing precautions. Water in accordance with ASTM D 1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Appearance. Visually examine the specimen for clarity, color, sediment and suspended matter.

4.2.4.2 Dilution. Dilute 100 mL of the specimen with 300 mL of water and maintain at 21° ± 1°C for 1 hour. No turbidity shall develop during the hour.

4.2.4.3 Assay. A 90C solidification point thermometer conforming to ASTM E 1 shall be used for this test. Fill a clean, dry test tube, approximately 1 inch in diameter, with sufficient quantity of specimen to reach the immersion mark on the thermometer when the bulb of the thermometer is approximately 1/2 inch above the bottom of the test tube. Place the test tube in an ice-water bath and cool without stirring to about 3°C below the assumed freezing point. Remove from the ice bath, stir to induce freezing, and record the point at which the temperature remains constant. (The temperature rises quickly to a temperature which is constant for approximately 30 seconds.) This temperature is taken as the freezing point.

4.2.5 Container leakage test. Place the container in each of the following positions, and leave it in each for a period of 15 minutes.

(a) Upright
(b) Upside down
(c) On one side (or one quadrant)
(d) On one end (or second quadrant)
(e) On other side (or fourth quadrant)

Examine the container after each period for any evidence of leakage.

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5. PACKAGING

5.1 Preservation. Acetic acid shall be preserved level A, B or C as specified (see 6.2) in accordance with Department of Transportation (DOT) regulations and any other applicable regulatory requirements, and the general requirements of PPP-C-2020.

5.1.1 Level A.

5.1.1.1 Unit packing.

5.1.1.1.1 Five-pound (lb) quantity. A quantity of 5 lb (+1 or -0 ounce) of acetic acid shall be unit packed in a glass bottle. The bottle shall be centered, closure end uppermost, and cushioned in a metal can. The bottle shall conform to group A, class 1, type d, style 1, grade optional, closure A or R, outer seal A of PPP-C-186. There shall be no evidence of leakage of contents from the filled and closed bottle, prior to cushioning and enclosing, when tested as specified in 4.2.5. The bottle shall be enclosed and cushioned as specified for enclosure B in PPP-C-2020.

5.1.1.1.2 Four hundred fifty-lb quantity. A quantity of 450 (+5 or -0) lb of acetic acid shall be unit packed in a nominal 55-gallon composite drum. The drum shall consist of a DOT Specification 2SL liner within a DOT 6D steel overpack. The steel drum shall be cleaned, phosphatized and painted as specified for the drums of PPP-D-729, using color forest green. The drum shall be closed in accordance with the drum manufacturer's instructions. There shall be no evidence of leakage when the drum is tested as specified in 4.2.5.

5.1.1.2 Intermediate packing. Six 5-lb unit packs of acetic acid, unit packed as specified in 5.1.1.1.1, shall be intermediately packed in a close-fitting fiberboard box conforming to class weather-resistant, grade V3c of PPP-B-636. Motion of contents shall be prevented by inserting fiberboard pads where needed. Pads shall be formed from material conforming to class weather-resistant, grade V3c or W5c of PPP-F-320. The box shall be closed as specified for the closure of interior containers in PPP-B-636.

5.1.2 Level B.

5.1.2.1 Unit packing.

5.1.2.1.1 Five-lb quantity. A 5-lb quantity of acetic acid shall be unit packed level B in the same manner as level A except that a polyethylene bottle without overpack shall be used. The bottle shall conform to group A, class 2, style 1, grade optional with closure A or R and outer seal A of PPP-C-186.

5.1.2.1.2 Four hundred fifty-lb quantity. A 450-lb quantity of acetic acid shall be unit packed level B in the same manner as specified for level A except that the drum shall consist of a DOT 2U inner liner with a DOT 37M overpack.

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5.1.2.2 Intermediate packing. Five-lb unit packs of acetic acid shall be intermediately packed unit B in the same manner as specified for level A, except that the box shall conform to class domestic, variety SW or DW, minimum grade 200 of PPP-B-636, and the fiberboard pads shall conform to class domestic, variety SW or DW, minimum grade 200 of PPP-F-320.

5.1.3 Level C.

5.1.3.1 Five-lb quantity. A quantity of 5 lb (+1 or -0 ounce) of acetic acid shall be preserved level C in a polyethylene bottle conforming to DOT Specification 2E.

5.1.3.2 Four hundred fifty-lb quantity. A quantity of 450 (+5 or -0) lb of acetic acid shall be preserved level C in a composite drum as specified for level B except that the phosphatizing shall not be required and the painting shall be in accordance with acceptable commercial practice.

5.2 Packing. Acetic acid shall be packed level A, B or C as specified (see 6.2) and in accordance with DOT regulations and any other applicable regulatory requirements.

5.2.1 Level A.

5.2.1.1 Five-lb quantity. Two intermediate packs of the 5-lb quantity of acetic acid shall be packed in a close-fitting wood box conforming to class 3, style optional of PPP-B-585. Motion of contents shall be prevented by inserting fiberboard pads where needed. Pads shall be formed from material conforming to minimum grade W6c of PPP-F-320. The box shall be closed as specified in PPP-B-585.

5.2.1.2 Four hundred fifty-lb quantity. The 450-lb quantity, preserved as specified in 5.1, shall require no further protection for shipment.

5.2.2 Level B.

5.2.2.1 Five-lb quantity. The 5-lb quantity shall be packed level B in the same manner as specified for level A, except that the box shall conform to class weather-resistant, grade V3c of PPP-B-636, and the box shall be reinforced by strapping or taping as specified therein.

5.2.2.2 Four hundred fifty-lb quantity. The 450-lb quantity, preserved as specified in 5.1.2.1.2, shall require no further protection for shipment.

5.2.3 Level C.

5.2.3.1 Five-lb quantity. The 5-lb quantity shall be packed level C in a manner to assure carrier acceptance and in accordance with Uniform Freight Classification Rules, National Motor Freight Classification Rules and the rules

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of any other intended mode of transportation. Containers shall be capable of protecting contents from supply source to initial destination and for a minimum duration of six months.

5.2.3.2 Four hundred fifty-lb quantity. The 450-lb quantity, preserved as specified in 5.1.3.2, shall require no further protection for shipment.

5.2.4 Unitization. Level A and B packs of the 5-lb quantity of acetic acid shall be palletized in accordance with the requirements for load type I of MIL-STD-147 using the soft wood pallet conforming to type IV of NN-P-71. The 450-lb quantity shall require no palletization. Level C packs of the 5-lb quantity shall be unitized in a manner to assure carrier acceptance and stable stackability in shipment and storage for a minimum duration of six months.

5.3 Marking. Containers and pallet loads of acetic acid shall be marked in accordance with MIL-STD-129 for military activities or in accordance with Fed. Std. No. 123 for civil agencies and in accordance with all applicable regulatory requirements. Each container shall be marked to show lot or batch number and date of manufacture of the acetic acid. In addition, each container shall be durably and legibly marked with the following precautionary information:

DANGER! CAUSES SEVERE BURNS

Use with adequate ventilation.
Keep away from heat and open flame.
Do not get liquid or vapor in eyes, on skin, or on clothing.
In case of contact, immediately flush skin or eyes with plenty of water for at least 15 minutes; for eyes, get medical attention.
Store in area maintained above 62°F.
If frozen, thaw by carefully moving container to warm area.

6. NOTES

6.1 Intended use. Glacial acetic acid is intended for use as a general neutralizing and acidifying agent and as a solvent for essential oils, resins, and other materials.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification,
(b) Level of preservation and packing required (see 5.1 and 5.2).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

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6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

6.5 Sampling and testing precautions. This specification covers inspection of chemical material which is potentially hazardous to personnel. All applicable safety rules, regulations and procedures must be followed in the handling and processing of this material.

MILITARY INTERESTS:

Custodians:

Army - EA
Air Force - 68

Review activities:

Army - MD, MI
DLA - GS

User activities:

Army - ER
Navy - AS, OS

CIVIL AGENCY COORDINATING ACTIVITIES:

AGR-AMS
GSA-FSS
HHS-NIH

Preparing activity:

Army - EA

Project No. 6810-B428

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
O-A-76E	ACETIC ACID, GLACIAL, TECHNICAL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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